

Abstract

This document specifies the **gen.3** protocol for neural track extrapolation in the LHCb experiment. Gen-3 is a targeted iteration on the Gen-2 best-of-family models with a single overriding design priority: *delivering an extrapolator binary that passes the Allen HLT1 standalone smoke test*. The Gen-2 audit ([gen2_new_data_results.tex](#), §5) showed that all eleven Gen-2 models fail Allen deployment, not because of architectural limits, but because of three data-generation bugs (C1–C3) and two model-level defects (A2, A4) that propagation accuracy numbers alone cannot detect. Gen-3 freezes three model candidates — one per family — fixes the five blocking issues in dependency order, and adds four theoretical improvements derived directly from the Gen-2 SciFi failure diagnostics. The protocol specifies the data regeneration, the training changes, the Allen-side validation sequence, the hard gates each candidate must clear, and the milestones that gate deployment.

Experimental Protocol: Third-Generation Neural Track Extrapolation for the LHCb Allen HLT1 Trigger Allen-first iteration on Gen-2 best-of-family models

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1 Introduction and priority statement

Gen-1 established that a [64, 64] MLP extrapolator can achieve 0.49 mm mean position error within the 64 kB CUDA `__constant__` memory budget. Gen-2 extended this to five MLP sizes, three PINN_v2 configurations, and three NeuralRK4 configurations on a new 50e6-track dataset with $\Delta z \in [25, 10000]$ mm, and performed a full detector-level characterisation against the VELO, UT and SciFi stations.

The Gen-2 conclusion is unambiguous: **the bottleneck is not the neural architecture**. The best Gen-2 candidate (`nrk4_small_1step`) already achieves 0.125 mm mean position error with 17.9 kB of weights. What blocks deployment is

1. the training dataset uses a *qop* convention incompatible with Allen (C1),
2. the training dataset does not cover the signed- Δz and wide input-coverage regimes that the Allen Kalman filter actually queries (C2, C3), and
3. the best-accuracy candidate either overshoots the memory budget (A3) or carries an uncorrected deterministic q/p -linear error at the SciFi stations (A2).

Gen-3 priority. Allen HLT1 deployability is the *primary* design variable; propagation accuracy is secondary. Every protocol decision below is justified against the Allen `ML_research/README.md` convention requirements and the three smoke-test gates, not against mean position error.

1.1 Research questions

RQ1 — Deployment. After fixing the five blocking issues identified in Gen-2, does at least one best-of-family model pass the three Allen smoke-test gates ($\chi^2/\text{ndof} < 2.0$, $|\delta p/p| < 1\%$, $\sigma(\text{pull}_x) < 1.5$) when connected to the standalone Kalman filter?

RQ2 — Which family wins in-situ. Which of {NeuralRK4, PINN_v2, MLP} generalises best to the in-situ Allen distribution (signed Δz , ± 3200 mm inputs, Allen *qop*), as opposed to the restricted distribution the Gen-2 models were trained on?

RQ3 — Theoretical improvements. Do the four targeted theoretical changes (Sec. 5) — (i) fixing the PINN_v2 q/p -linear magnet-kick bug, (ii) adding z_{start} to the input, (iii) detector- σ -weighted loss, (iv) optional Jacobian regularisation against finite-difference RK4 — each individually reduce at least one of the Gate-1/2/3 metrics on the same model?

2 Allen deployment constraints (hard gates)

All constraints in this section come from `Allen/ML_research/README.md` and `Allen/ML_research/gen2_true_da`. They are *mandatory*; a Gen-3 candidate does not “partially satisfy” them.

2.1 Conventions (C1–C3)

C1 — *qop* convention. The feature-space variable the model consumes must be

$$qop = c_{\text{light}} \cdot q/p\text{MeV}, \quad c_{\text{light}} = 299.792458 \text{ mm ns}^{-1} \text{ eplus}^{-1}. \quad (1)$$

This is Allen’s canonical unit; a factor of ~ 300 larger than the Gen-2 training convention. The fix is in the data generator; no runtime conversion inside the model loader is permitted.

C2 — signed Δz . The Allen Kalman filter performs forward *and backward* passes. The training distribution must therefore cover both signs:

$$\Delta z \sim \text{sign} \cdot \text{LogU}(25, 10000) \text{ mm}, \quad \text{sign} \sim \{-1, +1\} \text{ (balanced)}. \quad (2)$$

Log-uniform in $|\Delta z|$ over-represents the short VELO inter-sensor steps (25–30 mm).

C3 — input coverage. Inference-time inputs reach ± 3200 mm after the first forward Kalman update; training must cover at least this range. Two acceptable strategies: (a) harvest $(x_{\text{in}}, y_{\text{in}})$ pairs from a real Allen reconstruction at every layer interface, or (b) uniform sampling over the detector envelope $|x| \leq 3500$ mm, $|y| \leq 2500$ mm. Strategy (a) is preferred because it preserves the joint density on $(x, y, t_x, t_y, q/p)$.

2.2 Allen binary contract (A-series)

A3 — memory budget. Parameters + biases must fit in 64 kB CUDA `--constant--` memory, i.e. ≤ 16384 `float32` scalars. Quantisation to `float16` doubles the budget but must be preceded by a numerical-stability check (Sec. 8).

A4 — Jacobian correctness. The 5×5 Jacobian $\mathbf{J} = \partial \mathbf{y}_f / \partial \mathbf{y}_0$ with $\mathbf{y} = (x, y, t_x, t_y, qop)$ (Allen Kalman state; the z_0 and Δz inputs of Fix H are book-keeping variables and are *not* differentiated) must agree element-wise with a finite-difference RK4 reference. Per-input FD step sizes follow Allen’s own `compute_jacobian` convention $\epsilon = (10^{-3} \text{ mm}, 10^{-3} \text{ mm}, 10^{-6}, 10^{-6}, \max(|qop|, 10^{-4}, 10^{-8}))$. Structural-zero elements (where $|\mathbf{J}_{ij}^{\text{RK4}}| < \epsilon_i^{\text{abs}}$, with $\epsilon^{\text{abs}} = (10^{-6}, 10^{-6}, 10^{-8}, 10^{-8}, 10^{-10})$ for the five output rows) are scored on absolute error $|\mathbf{J}^\theta - \mathbf{J}^{\text{RK4}}| < 3\epsilon_i^{\text{abs}}$; the remainder are scored on relative error. Median relative error must be $< 10\%$ and 99th-percentile $< 30\%$. Frobenius-norm agreement is *not* sufficient (Gen-1/Gen-2 lesson).

A5 — output dimension. The Allen `MLPExtrapolator.cuh` V2 format expects *five* outputs $(x_f, y_f, t_{x,f}, t_{y,f}, qop_f)$. Gen-2 models output four. Gen-3 candidates must emit the fifth component; for `NeuralRK4` this is identity (*qop* conserved in vacuum), for `PINN_v2` and `MLP` it is an additional regression head.

A6 — metadata block. Each exported binary carries a provenance block: `qop_convention`, `c_light_value`, `dz_signed`, `feature_order`, `polarity` (MagDown only in baseline Gen-3), `loader_version_required`, `data_sha256`, `git_hash`, `training_seed`. Mismatch against the Allen Kalman state at load time is a hard error.

A7 — loader V3 patch. The existing Allen loader `MLPExtrapolator.cuh` hard-codes `float input[6]` and `float output[4]` in `propagate()` / `propagate_multistep()`; it reads `input_size` from the binary header and so does *not* crash on 7-input / 5-output models, but silently reads uninitialised stack memory for the 7th input and silently discards

the 5th output. Gen-3 therefore delivers a V3 patch to `MLPExtrapolator.cuh` (inputs $(x, y, t_x, t_y, qop, z_0, \Delta z)$ in that order; outputs $(x_f, y_f, t_{x,f}, t_{y,f}, qop_f)$) *as part of the milestone M0.5*. A binary advertising `loader_version_required=V3` refuses to load on a V2-only Allen checkout.

2.3 Smoke-test gates (G1–G3)

Defined in `ML_research/README.md`; baseline is the production RKN extrapolator.

Gate	Metric	Threshold
G1	$\langle \chi^2/\text{ndof} \rangle$ over standalone Kalman fit	< 2.0
G2	$ \langle p_{\text{fit}}/p_{\text{true}} - 1 \rangle $	$< 1\%$
G3a	$\sigma(\text{pull}_x)$	< 1.5
G3b	$ \langle \text{pull}_x \rangle $	< 0.1
RKN baseline	$(0.95 \pm 0.20, -0.04\%, 0.95)$	PASS

G3 is split into G3a (width) and G3b (bias): a model with $\langle \text{pull}_x \rangle = -2$, $\sigma = 1.4$ would pass the width-only version of G3 but is clearly mis-calibrated. G3b catches this.

3 Selected best-of-family candidates

Gen-3 iterates on three model candidates drawn from the Gen-2 detector-level audit (`gen2_new_data_results.tex` §4.5, §5). Rationale for each:

Primary: nrk4_tiny_1step. 4869 params, 19.5 kB (*fits A3 with headroom*), $\rho(\Delta x, q/p) = +0.02$ at SciFi T3 (*passes A2*), $|\Delta x| \approx 0.13$ mm one-shot. The *only* Gen-2 model that satisfies both A2 and A3 without further intervention; first candidate to attempt the smoke test.

Accuracy ceiling: nrk4_small_1step. 17925 params, 71.7 kB (*violates A3 by 12%*), $\rho = +0.03$ (*passes A2*), $4.9\ \mu\text{m}$ at VELO exit, $25.3\ \mu\text{m}$ at UT entry. Deployment via `float16` quantisation (35.9 kB); provides an upper bound on what the NeuralRK4 family can deliver.

PINN representative: pinn_v2_1am0.1. 68612 params, best Gen-2 SciFi position accuracy. Carries the $\rho(\Delta x, q/p) = +0.92$ magnet-kick bug (diagnostic ii of Gen-2 §4.5.1); blocked on A2 until the PDE residual is re-audited. Kept in Gen-3 to test whether fixing the physics-residual formulation (Fix G below) can rescue the physics-informed route.

Reference only: mlp_medium. 101k params, best Gen-2 MLP on validation loss but fails A3 by a large factor. Retained as a no-physics control; not a deployment target.

The legacy RK-PINN, legacy PINN, and the remaining four MLP sizes are *not carried forward* — Gen-2 already showed they are dominated by the four above on at least one of the three Allen gates.

4 Data improvements: fixing C1–C3 and A5

The Gen-3 training dataset `train_50M_gen3.npz` is regenerated from scratch. Changes relative to `train_50M_dz25.npz`:

4.1 Generator patches

1. **C1 — *qop* convention.** Replace the scalar computation in `utils/rk4.propagator.py`:

```
# old
qop = charge / p_MeV
# new (Gaudi mixed units: (mm ns-1 eplus) / MeV)
qop = 2.99792458e+2 * charge / p_MeV # ~ = 3e-2 at p = 10 GeV
```

Sanity arithmetic: $c_{\text{light}}/10000 \text{ MeV} = 299.792458/10\,000 \approx 3.00 \times 10^{-2}$, matching the `ML_research/README.md` worked example. The previous draft coefficient `2.99792458e-1` was a factor 10^3 too small — worse than the Gen-2 convention. **Companion edit in the integrator:** replace `C_LIGHT = 2.99792458e-4` in `utils/magnetic_field.py` with `C_LIGHT = 1.0e-6` so that $\kappa = c_{\text{light}} \cdot qop$ evaluates to the *same* number as the Gen-2 expression $\kappa = (2.99792458 \times 10^{-4}) \cdot (q/p_{\text{MeV}})$, i.e. c_{light} is absorbed into qop exactly once. **Regression test (closed-form).** Uniform $B_y = -1 \text{ T}$, $B_x = B_z = 0$, $q = +1$, $p = 10 \text{ GeV}$, $\Delta z = 1000 \text{ mm}$, initial $(x_0, y_0, t_{x,0}, t_{y,0}) = (0, 0, 0, 0)$: the analytic trajectory is a circular arc of radius $R = p/(cB) = 33.356 \text{ m}$ giving $x_f = R(1 - \cos(\Delta z/R)) = 14.990 \mu\text{m}$ to six figures. Pass tolerance $|\Delta x_f| < 1 \mu\text{m}$ at this test point, spot-checked on 10k tracks. A second closed-form test with $B_x = +1 \text{ T}$ (everything else zero) isolates the B_x branch of the Lorentz RHS — required because Fix G depends on all three field components being correctly evaluated.

2. **C2 — signed Δz .** Replace $\Delta z \sim \mathcal{U}(25, 10000)$ with

$$\Delta z = \epsilon \cdot 10^u, \quad \epsilon \sim \{-1, +1\}, \quad u \sim \mathcal{U}(\log_{10} 25, \log_{10} 10000). \quad (3)$$

The RK4 integrator handles negative Δz by stepping with $-h$; the magnetic-field module is independent of step sign.

3. **C3 — full input coverage.** Extend the sampling envelope to $x_0 \sim \mathcal{U}(-3500, +3500) \text{ mm}$, $y_0 \sim \mathcal{U}(-2500, +2500) \text{ mm}$, and widen the slopes to $t_{x,0} \sim \mathcal{U}(-0.4, 0.4)$, $t_{y,0} \sim \mathcal{U}(-0.35, 0.35)$ to accommodate Kalman states that extrapolate *off-axis and wide-angle* simultaneously. Retain the $|x|, |y| < 5000 \text{ mm}$, $|t| < 2.0$ post-integration rejection. The log-momentum range $p \in [1, 200] \text{ GeV}/c$ is unchanged.
4. **A5 — fifth output.** Add qop_f to the output array $\mathbf{Y} \in \mathbb{R}^{N \times 5}$. For vacuum propagation $qop_f \equiv qop_0$; this column is kept explicitly so the Allen binary contract is honoured without a loader-side synthetic column.

4.2 Data provenance hardening

Alongside the regenerated `.npz` file, write `train_50M_gen3.meta.json` capturing `qop_convention=allen_v1`, `dz_signed=True`, `x_range=[-3500, 3500]`, `generator_git_hash`, `field_map_sha256`, `batch_seeds={42 + 7919 i}`, `n_batches`, `created_utc`. This metadata is passed through to the Allen binary's A6 block on export.

4.3 Optional: realism extensions (deferred to Gen-3.1)

The following are *not* in the baseline Gen-3 dataset but are staged for Gen-3.1 once the Allen smoke test passes on the vacuum dataset:

- Multiple scattering: Highland formula applied as a random slope kick at each RK4 sub-step.

- Energy loss: Bethe–Bloch dE/dx applied between sub-steps; makes $qop_f \neq qop_0$ and activates the A5 output.
- Mixed magnet polarity (MagUp via $B_y \rightarrow -B_y$).

5 Theoretical improvements

Each item is derived from a specific diagnostic in Gen-2 §4.5.1 and §5. Numbering continues from the Gen-2 fix labels (A–F).

5.1 Fix G — PINN_v2 PDE residual re-audit (A2)

Gen-2 measured $\rho(\Delta x, qop) = +0.92, +0.92, +0.99$ for PINN_v2 with $\lambda \in \{0.1, 1.0, 10.0\}$. The correlation *grows with* λ , which rules out a pure data-fit explanation and implicates the PDE residual itself. Two candidate causes:

1. The field-interpolation module returns only B_y , and the Lorentz RHS (Eqs. ??–?? of Gen-2 protocol) is evaluated with $B_x = B_z = 0$ by default. The measured field map has $|B_x|$ up to 0.15 T in the fringe regions that the SciFi-station tracks traverse.
2. The non-dimensionalisation factor $\sigma_k/\Delta z$ (Eq. 5.9 of Gen-2 protocol) does not remove the q/p dependence of the residual scale, so the penalty is effectively weighted by $|q/p|$.

Action: (i) restore all three field components in `models/pinn_v2.compute_physics_loss`; (ii) unit-test the PDE residual against the analytic uniform- B_y trajectory (closed-form circular arc) to confirm the residual is identically zero on the analytic ground truth. Pass criterion: $\rho(\Delta x, qop) < 0.1$ at SciFi T3 on the Gen-3 test split.

5.2 Fix H — add z_{start} to the input

The field map is strongly z -dependent ($|B_y|$ peaks at $z \simeq 5007 \text{ mm}$) but the Gen-2 input vector $\mathbf{x} = (x_0, y_0, t_{x,0}, t_{y,0}, qop, \Delta z)$ does *not* carry z_0 . All Gen-2 models implicitly assume $z_0 = 0$; the RK4 ground truth does not. This produces a deterministic z_0 -dependent residual visible in the per-station error growth (Gen-2 Tab. ??). Gen-3 extends the input to $\mathbf{x} \in \mathbb{R}^7$:

$$\mathbf{x} = (x_0, y_0, t_{x,0}, t_{y,0}, qop, z_0, \Delta z). \quad (4)$$

The z_0 column already exists in the Gen-2 $\mathbf{Z}$ array; the regenerated Gen-3 dataset includes it in $\mathbf{X}$ directly. Cost: one extra first-layer weight column per neuron; negligible memory impact.

5.3 Fix I — detector- σ -weighted endpoint loss

Gen-2 uses a plain component-weighted MSE (Eq. 5.7 of Gen-2 protocol) with weights $1/\sigma_k$ taken from the *training-set* standard deviation. This produces uniform tolerance across all Δz and all momenta. But the downstream Kalman filter evaluates residuals in *detector-resolution units*: $(\sigma_x^{\text{VELO}}, \sigma_y^{\text{VELO}}) = (12 \mu\text{m}, 12 \mu\text{m})$, $(\sigma_x^{\text{UT}}, \sigma_y^{\text{UT}}) = (50 \mu\text{m}, 5 \mu\text{m})$ (stereo UT layers; y -strips), $(\sigma_x^{\text{SciFi}}, \sigma_y^{\text{SciFi}}) = (60 \mu\text{m}, 500 \mu\text{m})$ (stereo SciFi layers). Slope tolerances are set by differentiating position resolution over typical inter-station lever arms: $\sigma_{t_x} = 1e - 4$ (VELO/UT), $3e - 5$ (UT→SciFi), σ_{t_y} identical scale. Gen-3 replaces the training loss with

$$\mathcal{L}_{\text{det}} = \frac{1}{N} \sum_i \sum_{k \in \{x, y, t_x, t_y\}} \left(\frac{\hat{y}_{ik} - y_{ik}}{\sigma_k^{\text{det}}(z_f^i)} \right)^2, \quad (5)$$

where $\sigma_k^{\text{det}}(z)$ is the detector resolution of the *nearest station* to z_f . This directly couples the training objective to the quantity Allen Gate G3 measures. Expected effect: pull-width narrowing at VELO and UT at the cost of mild degradation in the SciFi-interior region where no station provides a hit.

5.4 Fix J — Jacobian regularisation via finite-diff RK4 (A4)

Gen-2 §6 (Jacobian Frobenius norms) showed all models sit within a factor of 2 of the RK4 reference on $\|\mathbf{J}\|_F$, but this is a scalar summary; element-wise Jacobian agreement was never measured. The Kalman covariance update is linear in the Jacobian, so an element-wise bias of even 20 % can inflate $\sigma(\text{pull}_x)$ past Gate G3. Gen-3 adds an optional regulariser:

$$\mathcal{L}_{\text{Jac}} = \lambda_J \cdot \left\| \frac{\partial f_{\boldsymbol{\theta}}}{\partial \mathbf{x}_0} - \frac{\partial f_{\text{RK4}}}{\partial \mathbf{x}_0} \right\|_F^2, \quad (6)$$

with $\partial f_{\boldsymbol{\theta}}/\partial \mathbf{x}_0$ computed by `torch.func.jacfwd` and the RK4 Jacobian by centred finite differences using Allen’s per-input step sizes $\epsilon = (10^{-3}, 10^{-3}, 10^{-6}, 10^{-6}, \max(|qop| \cdot 10^{-4}, 10^{-8}))$ (*not* a flat 10^{-4} , which would be noise-dominated on the slope and *qop* rows). Activated on a 5e3-sample mini-batch every 10 optimiser steps to amortise the finite-difference cost. λ_J swept in $\{0, 10^{-3}, 10^{-2}, 10^{-1}\}$. The 10-step amortisation cadence is a throughput/signal trade-off: every optimiser step logs $\mathcal{L}_{\text{Jac}}$ (even on steps where it is not applied) via a cached previous-value so that post-hoc diagnostic plots can detect drift between the data gradient and the Jacobian gradient.

5.5 Fix K — optional: predict residual to straight-line ansatz

For the MLP family specifically, the PINN_v2 envelope trick (prediction = straight line + learnable correction) provides a 10 %-level accuracy improvement when the straight-line baseline already captures the ballistic component. Gen-3 adds an ablation in which `mlp_medium` is trained to emit $\boldsymbol{\delta} = \hat{\mathbf{y}} - \mathbf{y}_{\text{base}}$ instead of $\hat{\mathbf{y}}$ directly. This is a pure output-space transformation; no architectural change.

5.6 Fix L — optional: non-uniform multi-step NeuralRK4

`nrk4_small1_2step` underperformed `1step` on SciFi ($\rho = -0.49$); the suspected cause is that the step split is uniform in Δz , so both RK4 sub-steps see similar field gradients. A non-uniform split with an internal breakpoint at the magnet peak $z \simeq 5000 \text{ mm}$ is staged as a Gen-3.1 ablation if `1step` fails at the SciFi-interior.

6 Model-level changes per candidate

Model	Gen-3 architectural change	Gen-3 loss change	New params
<code>nrk4_tiny_1step</code>	$\mathbf{x} \in \mathbb{R}^7$ (Fix H); 5-output head (A5).	Fix I; Fix J at $\lambda_J \in \{0, 10^{-3}, 10^{-2}\}$.	≈ 5000 ; 20 kB.
<code>nrk4_small1_1step</code>	As above + float16 export.	Fix I; Fix J.	$\approx 18\text{k}$; 36 kB (fp16).
<code>pinn_v2_lam0.1</code>	Fix G (re-audited PDE residual + full $\mathbf{B}$); Fix H; 5-output head.	Fix I; Fix G (unit-tested residual); λ_{pde} swept $\{0.03, 0.1, 0.3\}$ only.	$\approx 69\text{k}$; <i>reference only</i> .
<code>mlp_medium</code>	Fix H; 5-output head; Fix K ablation.	Fix I.	$\approx 101\text{k}$; <i>reference only</i> .

7 Training protocol

Largely inherited from Gen-2 §6. Explicit Gen-3 deltas:

- **Split.** 80/10/10 on the regenerated dataset, seed 42, stratified by $\text{sign}(\Delta z)$ so that the backward pass is represented in validation and test.

- **Optimiser.** AdamW, $\beta = (0.9, 0.999)$, weight decay 10^{-4} . Base learning rate 10^{-3} for NeuralRK4/MLP, 5×10^{-4} for PINN_v2. Cosine annealing, 5-epoch linear warm-up, max 200 epochs, early stop on validation loss with patience 30 (NeuralRK4/MLP) / 40 (PINN_v2).
- **Batch size.** 2048 for NeuralRK4/MLP, 1024 for PINN_v2, reduced to 512 on mini-batches that evaluate Fix J.
- **Seeds.** Global seed 42; dataset seed 42; per-batch seeds $s_i = 42 + 7919i$ (unchanged from Gen-2).
- **Infrastructure.** Nikhef HTCondor cluster; `condor/submit_gen3.sub`. MLflow experiment `gen_3_track_extrapolation`.

8 Validation protocol

Gen-3 validation is a three-stage funnel. A candidate that fails any earlier stage does not proceed to the next.

8.1 Stage 1 — propagation-level sanity (reuse Gen-2 characterisation)

On the Gen-3 test split, compute the Gen-2 detector-level metrics: per-station $\langle |\Delta x| \rangle$, $\rho(\Delta x, qop)$ at SciFi T3, per-leg residual growth, two-dimensional residual heatmap in $(\Delta z, p)$. Pass criteria:

- VELO exit: $\langle |\Delta x| \rangle < 12 \mu m$ (VELO σ).
- UT entry: $\langle |\Delta x| \rangle < 50 \mu m$ (UT σ).
- SciFi T3: $|\rho(\Delta x, qop)| < 0.1$ for PINN_v2; < 0.3 otherwise.
- Backward pass ($\Delta z < 0$) metrics within a factor 2 of forward.

8.2 Stage 2 — Jacobian correctness (A4)

On 1000 randomly selected test samples, compute the 5×5 Jacobian $\mathbf{J}_\theta$ via `torch.func.jacfwd` over the Allen Kalman state (x, y, t_x, t_y, qop) only (not $z_0, \Delta z$) and $\mathbf{J}_{RK4}$ by centred finite differences with Allen’s per-input step sizes (A4). Pass criterion per A4: median element-wise relative error $< 10\%$, 99th-percentile $< 30\%$ on the non-structural-zero elements; structural zeros scored on absolute error. Report $(\mathbf{J}_\theta - \mathbf{J}_{RK4})/\mathbf{J}_{RK4}$ per matrix element so that any systematic bias is visible.

8.3 Stage 2.5 — exported binary round-trip

Load the exported `.bin` into a reference `numpy`-only loader (`scripts/load_allen_v3_ref.py`) and evaluate 100 test tracks. Assert bitwise agreement ($|\Delta| < 10^{-6}$ relative) with the PyTorch forward pass on the same inputs. This catches the class of failure where Stage 3 regresses for loader reasons rather than model reasons — the single most common Gen-1/Gen-2 deployment bug.

8.4 Stage 3 — Allen standalone smoke test (G1–G3)

Export the model to the `MLPExtrapolator.cuh` V2 binary via `scripts/export_allen_v2.py` and run `ML_research/standalone/run_extrapolators` against the RKN baseline. Record G1/G2/G3 on the same sample of $\mathcal{O}(10^5)$ tracks. A candidate is declared *deployment ready* if and only if all three gates pass simultaneously.

8.5 Validation harness outputs

Each candidate produces: `gen3_candidate_{name}_stage1.csv` (per-station metrics), `..._stage2.csv` (Jacobian element-wise stats), `..._stage3_smoketest.log` (Allen harness stdout), `...bin` (exported weights), and `...meta.json` (A6 provenance block).

9 Iteration plan

Dependency-ordered; later milestones assume earlier ones pass.

1. **M0 — data regeneration.** Patch generator for C1/C2/C3/A5, regenerate 50e6 tracks (`condor/submit_gen3_datagen.sub`); verify closed-form uniform- B_y and uniform- B_x tests (§4) with tolerance $|\Delta x_f| < 1 \mu m$ at $(p, \Delta z) = (10 \text{ GeV}, 1 \text{ m})$. *Gate: dataset SHA-256 recorded; both closed-form tests pass; spot-check 10k tracks match the Allen-convention reference.*
2. **M0.5 — Allen loader V3 patch (A7).** Patch `MLPExtrapolator.cuh propagate()` / `propagate_multistep()` to accept the 7-input / 5-output format; add the `loader_version_required` header field; add `compute_jacobian` extension to the 5×5 Kalman-state Jacobian. *Gate: unit test loading a Gen-3 placeholder binary into a local Allen build succeeds; reference Python loader (`scripts/load_allen_v3_ref.py`) and C++ loader produce bitwise-identical outputs on 100 test tracks.*
3. **M1 — baseline re-training.** Train `nrk4_tiny_1step` with Fixes H+I only (no J, no G). *Gate: Stage 1 passes.*
4. **M2 — Jacobian validation.** Run Stage 2 on the M1 model. If median element-wise error $> 10\%$, add Fix J (Jacobian regulariser) and retrain. *Gate: Stage 2 passes.*
5. **M3 — first smoke test.** Export M2 model and run Stage 3. *Gate: at least one of G1/G2/G3 is closer to baseline than the Gen-2 true-data result (Tab. ?? in Gen-2 report).*
6. **M4 — smoke-test pass.** Iterate on λ_J and hidden-dim choice until all three gates pass. *Gate: $G1 < 2.0$, $|G2| < 1\%$, $G3 < 1.5$ simultaneously.*
7. **M5 — accuracy ceiling.** Repeat M1–M4 for `nrk4_small_1step` with float16 export. *Gate: same three gates pass at higher accuracy margin.*
8. **M6 — PINN_v2 rescue.** Train `pinn_v2_lam0.1` with Fix G + Fix H + full **B**. *Gate: $\rho(\Delta x, qop) < 0.1$; if not, PINN_v2 is retired.*
9. **M7 — Kalman integration.** Replace the RKN extrapolator in the Allen HLT1 sequence with the M4/M5 model; record event-throughput impact and end-to-end physics performance (tracking efficiency, fake rate, momentum resolution). *Gate: no regression in tracking efficiency; per-track inference time $\leq$ RKN baseline on the target HLT1 GPU (A5000 assumed; re-state on the actual deployment GPU if different); throughput improvement measurable at the event level.*

10 Known limitations carried forward

- **Material effects — confounder for G2.** Baseline Gen-3 trains on vacuum propagation; Bethe–Bloch and Highland are staged for Gen-3.1. This is a test-design confounder, not just a limitation: a failing G2 at M4 cannot be attributed to the extrapolator versus missing dE/dx . The mitigation adopted here is to run Stage 3 with *material effects disabled in the*

standalone Kalman filter at M_4 , re-enabling them only at Gen-3.1. A light-weight Bethe–Bloch correction in the generator (dE/dx between RK4 sub-steps, makes $qop_f \neq qop_0$ and exercises the A5 output head) is kept as a fallback if Stage 3 with material-off indicates residual $\delta p/p$ sensitivity.

- **Single polarity, symmetry assumption.** MagDown only; MagUp would be exact under $qop \rightarrow -qop$ only for a pure B_y field. With the real interpolated field map (B_x, B_z non-zero in fringes) the symmetry is approximate. MagUp training is staged for Gen-3.1; the A6 polarity header field ensures a MagUp Allen run loading a MagDown binary raises a hard error.
- **Field-map uncertainty.** Gen-3 treats the `twodip.rtf` field map as ground truth; its measured RMS error is $\sim 1.3\%$. A 1% field error maps directly to a $\sim 1\%$ $\delta p/p$ bias in G2 regardless of extrapolator quality; Gate G2 should be read modulo this irreducible floor.
- **Slope-rejection cut.** The Gen-2 post-integration rejection $|t| < 2.0$ is retained but tightened to $|t| < 0.5$, comfortably above the LHCb geometrical acceptance of ~ 0.3 rad and the training envelope of $\pm 0.4/0.35$.
- **No measurement noise.** The data generator emits exact RK4 trajectories; Allen feeds Kalman-estimated states with covariance. The training-inference mismatch on input distribution is mitigated by C3 (wide sampling) but not eliminated.
- **Fix J cost.** Jacobian regulariser is expensive; per-epoch training time projected at $2\text{--}3\times$ baseline. Acceptable within the long-queue HTCondor policy.

11 Reproducibility

- **Environment.** TE conda env; PyTorch 2.9.1+cu128; Python 3.10; identical to Gen-2. `environment.yml` and `requirements.txt` pinned.
- **Code.** Branch `gen_3` of `TrackExtrapolation`. New modules: `experiments/gen_3/utls/rk4_propagator` (C1-patched), `utls/magnetic_field.py` (shared with Gen-2), `models/architectures.py` (Fix H input shape, A5 output head), `models/train.py` (Fix I detector- σ loss, Fix J Jacobian regulariser).
- **Configs.** `experiments/gen_3/configs/` with one JSON per candidate; each declares `applied_fixes=[C1,C2,C3,A5,H,I,...]` and `supersedes={gen2_experiment_id}`.
- **MLflow.** Experiment `gen_3_track_extrapolation`; run tags include `applied_fixes`, `supersedes`, `loader_version`, `dataset_sha256`, `generator_git_hash`, `polarity`. Metrics (floats, not booleans) include `stage1_velo_mean_dx`, `stage1_ut_mean_dx`, `stage1_scifi_rho_dx_qop`, `stage1_backward_pass_ratio`, `stage2_jac_median_rel_err`, `stage2_jac_p99_rel_err`, `smoketest_g1_ch`, `smoketest_g2_dpp_bias`, `smoketest_g3a_pull_std`, `smoketest_g3b_pull_mean`, `throughput_per_track`. Two passing candidates must be rank-orderable from MLflow alone.
- **Data.** `train_50M_gen3.npz + ....meta.json`. SHA-256 logged in MLflow and written into every exported Allen binary (A6).
- **Allen binaries.** Written to `Allen/ML_research/models/gen3/` with the filename pattern `{candidate}_{applied_fixes}_{git_hash}.bin`.

12 Summary

Gen-3 is a *deployment-first* iteration. The protocol does not introduce new model families; instead it takes the best Gen-2 candidate per family, patches the five blocking issues identified in the Gen-2 §5 Allen-readiness audit (C1, C2, C3, A2, A3), adds four theoretical improvements (Fixes G–J) derived from the Gen-2 §4.5.1 SciFi failure diagnostics, and runs each candidate through a three-stage validation funnel terminating in the Allen standalone smoke test. The primary success metric is binary: at least one candidate passes G1+G2+G3 simultaneously at milestone M4.

- **Primary candidate:** `nrk4_tiny_1step` with Fixes C1+C2+C3+A5+H+I; first to attempt the smoke test because it is the only Gen-2 model that satisfies A2 and A3 simultaneously.
- **Accuracy ceiling:** `nrk4_small_1step` with the same fixes plus float16 export.
- **Physics-informed rescue:** `pinn_v2_lam0.1` with Fix G; if the $\rho(\Delta x, qop)$ criterion is not met after Fix G, PINN_v2 is retired from the deployment path.
- **Reference only:** `mlp_medium` with Fix K ablation; not a deployment target.

A Gen-3.1 follow-up is staged to address material effects (Bethe–Bloch, Highland) and mixed magnet polarity once the vacuum deployment is proven.

Reviewer pass-through. This protocol incorporates blocking issues B1–B4 (C1 coefficient factor- 10^3 error, Allen loader V3 contract, Jacobian dimensionality, component-dependent detector σ) and advisable items N1–N15 from the **experiment-reviewer** audit (2026-04-23); all pre-M0 gates have been hardened to numerically enforceable thresholds.

References

- [1] G. Scriven, *Gen-1 neural track extrapolation for Allen HLT1*, internal note, 2026.
- [2] G. Scriven, *Gen-2 neural track extrapolation on the `train_50M_dz25` dataset*, internal note, `gen2_new_data_results.tex`, 2026.
- [3] G. Scriven, *Gen-2 experimental protocol*, internal note, `gen2_experimental_protocol.tex`, 2026.
- [4] LHCb Allen team, *ML_research: machine-learning extrapolator integration in Allen HLT1*, `Allen/ML_research/README.md`, 2026.
- [5] G. Scriven, *Gen-2 true-data Allen deep dive*, `Allen/ML_research/gen2_true_data_deep_dive.md`, 2026.